

Appendix 22.A Smart Antennas for CDMA

Smart antennas can also be used for the increase of the capacity in CDMA systems. While less relevant for modern systems (CDMA was mainly used in 3G cellular systems), it is still interesting to understand the operating mechanism. The way this capacity increase is achieved is subtly different from TDMA/FDMA systems discussed in Sec. 22.1. In a CDMA system, all UEs use the same carrier frequency, and are distinguished only by the spreading codes they employ. The suppression of users with other codes is not perfect, and a cell is judged to be "full" when the residual interference from all the other K users becomes comparable to the admissible SIR (refer to Chapter 19 for more details and a discussion of the assumptions):

$$SIR_{\text{threshold}} = \frac{P_{\text{desired}}}{\Sigma P_{\text{interfere}}} \quad (22.A.1)$$

where for the case of perfect power control, $\Sigma P_{\text{interfere}} = K \cdot P_{\text{interfere}}$. For typical (voice) CDMA systems, the number of users is on the order of 30. Thus, there is a large number of possible interferers, each of which contributes only a small part of the interference. It is thus *not* possible to suppress interference, since the number of interferers that can be suppressed is smaller than or equal to the number of antenna elements. Rather, the goal of the smart antenna is to enhance the received signal power. An N_{BS} -element antenna can increase the received signal power by a factor of N_{BS} , so that approximately

$$P_{\text{desired}} = M_c \cdot N_{\text{BS}} \cdot P_{\text{interfere}}, \quad (22.A.2)$$

where M_c is the spreading factor. In that case, the number of admissible users in the cell becomes

$$K = \frac{M_c \cdot N_{\text{BS}}}{SIR_{\text{threshold}}} \quad (22.A.3)$$

and thus increases linearly with the number of antenna elements. Note that this derivation is based on numerous simplifying assumptions, especially with respect to the channel model. In practice, the achieved capacity gain is considerably smaller.

The above considerations assumed that there is a large number of users in a CDMA system, and all of them contribute to the interference in a similar way. This assumption holds if there are mainly voice users. However, in third-generation networks, high-rate data transmission is considered an important application. Such a high-rate user creates considerably more interference, so that a suppression of such a user by placing a null in its direction becomes desirable. In other words, high-rate users lead to scenarios and results that are more similar to the SFIR approach than the traditional CDMA approach.

Example Consider a CDMA system with spreading factor 128 for voice users, and 4 for data users, and an SIR threshold of 9 dB. There is one data user. How many voice users can be accommodated without smart antennas? How many can be accommodated with a 2-element smart antenna?

Solution

The SIR threshold (on a linear scale) is 8. Without smart antennas, the capacity of the system (in units of voice users) can be obtained from

$$K = \frac{128}{8} = 16. \quad (22.A.4)$$

Since the spreading factor for data users is smaller by a factor of 32 compared to voice users, each data user contributes (approximately) 32 times the interference. The data users thus require a capacity of $K_{\text{data}} = 32$, so that $16 - 32 = 0$ voice users can be accommodated. Consider now the case with smart antennas. When the smart antennas are used solely for the enhancement of the desired signal power, the capacity of the cell (in units of voice users) becomes

$$K = \frac{128 \cdot 2}{8} = 32 \quad (22.A.5)$$

so that now $32 - 32 = 0$ voice users can be serviced. If, however, the smart antennas are used to suppress the data users, a higher capacity can be achieved: as we have seen in Section 12.4.3, we can suppress 1 interferer completely with 2 antennas, while retaining a diversity order of 1 for the desired signal. Thus, in our example, we can suppress the data users while retaining $N_r = 1$ for the voice users. The capacity for the voice users only thus becomes

$$\frac{128}{8} = 16 \quad (22.A.6)$$

and 16 voice users can be accommodated.

End of example